Review Article

Impact of Cancer and Its Treatments on Cognitive Function: Advances in Research From the Paris International Cognition and Cancer Task Force Symposium and Update Since 2012

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Abstract

Context. Although cognitive impairments have been identified in patients with non—central nervous system cancer, especially breast cancer, the respective roles of cancer and therapies, and the mechanisms involved in cognitive dysfunction remain unclear.

Objectives. To report a state-of-the-art update from the International Cognitive and Cancer Task Force conference held in 2012.

Methods. A report of the meeting and recent new perspectives are presented.

Results. Recent clinical data support that non—central nervous system cancer per se may be involved in cognitive dysfunctions associated with inflammation parameters. The role of chemotherapy on cognitive decline was confirmed in colorectal and testicular cancers. Whereas the impact of hormone therapy remains debatable, some studies support a negative impact of targeted therapies on cognition. Regarding interventions, preliminary results of cognitive rehabilitation showed encouraging results. The methodology of future longitudinal studies has to be optimized by a priori end points, the use of validated test batteries, and the inclusion of control groups. Comorbidities and aging are important factors to be taken into account in future studies. Preclinical studies in animal models highlighted the role of cancer itself on cognition and support the possible benefits of prevention/care during chemotherapy. Progress in neuroimaging will help specify neural processes affected by treatments.

Conclusion. Clinical data and animal models confirmed that chemotherapy induces direct cognitive deficit. The benefits of cognitive rehabilitation are still to be confirmed. Studies evaluating the mechanisms underlying cognitive impairments using advanced neuroimaging techniques integrating the evaluation of genetic factors are ongoing. J Pain Symptom Manage 2015;50:830—841. © *2015 American Academy of Hospice and Palliative Medicine. Published by Elsevier Inc. All rights reserved.*

Key Words

Cancer, chemotherapy, cognition, elderly, neuroimaging

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Accepted for publication: July 6, 2015.

0885-3924/$ - see front matter
http://dx.doi.org/10.1016/j.jpainsymman.2015.06.019

Introduction

It is now accepted that cancer treatments may induce cognitive impairments, particularly in the domains of memory, concentration, information processing speed, and executive functions. Cancer-related cognitive impairment was first reported in the context of brain irradiation. More recently, research largely focused on neurotoxicity associated with chemotherapy, leading to the general concept of "chemofog."[1] Most of these studies have been conducted in young patients with localized breast cancer, and it appears that a subgroup experiences, in most cases, subtle or moderate cognitive dysfunction.[2] The impact of hormone therapy on cognition is not well understood.[3,4]

Although it is acknowledged that chemotherapy is a risk factor for cognitive deficits, there are still many unresolved questions. These issues include the characterization of subgroups at risk, the impact of cancer per se and of other factors like aging or comorbidities, the specific effects of different treatment modalities, and the optimal way to improve the management of cognitive impairment in routine practice. Furthermore, there are still important gaps in our knowledge and understanding of the mechanisms implicated in cognitive disorders.

Over the past 10 years, the International Cognitive and Cancer Task Force (ICCTF) has organized a biannual multidisciplinary meeting involving neuropsychologists, clinical and experimental psychologists, medical oncologists, imaging experts, basic science researchers, nurses, and allied health professionals, with the aim of sharing state-of-the-art knowledge regarding the impact of cancer and its treatments on cognitive function. The first meetings mainly focused on cognitive changes secondary to adjuvant chemotherapy for breast cancer.[5,6] Based on a multidisciplinary working group approach, in 2011, the ICCTF published recommendations to harmonize the methodology of studies evaluating cognitive function in patients with cancer.[7] In 2012, the ICCTF symposium held in Paris aimed to update the advances in research on cognition in non—central nervous system (CNS) cancers. During this meeting, new aspects of research were presented, including important advances in basic science (various preclinical animal models) and in human brain imaging based on magnetic resonance imaging (MRI) and positron emission tomography (PET). In addition, translational research, including the role of different inflammatory biomarkers, was presented. The aim of this article is to summarize the main topics and advances presented during the symposium that have since been published and to review the new perspectives that have emerged since the Paris meeting.

Report From the 2012 ICCTF Paris Meeting

How to Improve Methodological Aspects of Cognitive Clinical Trials?

The development of well-designed clinical trials assessing cognitive function in patients with cancer was an important issue and long considered as critical by all clinical and neuropsychological professionals in the field. Past and current discussions focused on ways of improving some methodological aspects in studies assessing cognition. First, it was recommended to define an a priori cognitive end point and to use a validated neuropsychological battery of tests.[7] Thus, different methods of statistical analysis had to be debated. B. Small demonstrated that traditional statistical methods, such as those based on repeated analysis of variance, have important limitations, especially in the case of missing data and unequally spaced measurement intervals. He proposed alternative models more suitable for analysis of longitudinal studies evaluating cognitive function. Random effects models allow longitudinal data to be processed more flexibly and innovatively, with few penalties and less impact from missing data, and the ability to take into account practice effects.[8,9] Other models such as growth mixture models[10] allow testing hypotheses regarding the presence of subgroups of patients, some of whom may be experiencing a precipitous decline. Latent change score models enable investigators to link changes across multiple variables and to test hypotheses regarding whether changes in one aspect (e.g., biological markers) may precede or follow changes in another aspect (e.g., behavioral tasks).[11] Finally, a presentation of integrative data analysis[12,13] received considerable attention from scientists. Similar to a meta-analysis, but with raw data, information from multiple samples (which is often the situation in cognitive studies) could be combined into a single data analysis, despite the fact that all outcomes may not be measured using the same instruments. This type of analysis is expected to increase statistical power and generalizability of results.

Another debate, conducted by Collins et al.,[14] concerned the choice of appropriate control groups and led to the suggestion of choosing suitable published test-retest norms for all neuropsychological tests (that may require local controls) even if it remains a challenge. She showed significantly different rates of cognitive decline in patients treated with chemotherapy according to the control method used. In addition, the choice of local controls is also challenging. Healthy controls can be well matched for key demographic variables, but possible confounding factors such as constitutional risk factors, psychological distress, and disease-related cognitive changes, if not taken into account, may limit the analysis. Finally, control patients who did not receive the treatment of interest are an appropriate group. They match on the effects of the disease itself, host factors, and stress and distress associated with the diagnosis. However, these patients may receive other treatments and could then exhibit different characteristics of the disease. Nonetheless, the impact of the treatment itself on cognitive function remains difficult to assess. Because the choice of the control groups depends on the design of the study, it was suggested for future studies to use pooling of control data from harmonized studies to establish an appropriate normative database. Furthermore, the use of statistical matching algorithms, such as propensity score matching, may be a useful method to incorporate into future studies.[15]

What Is the Impact of Age and Comorbidities on Cognitive Decline?

A major focus of the Paris meeting was the impact of cancer and its treatments in elderly patients. Indeed, the number of elderly patients with cancer is rising, and the indications for adjuvant chemotherapy after surgery have become widespread in the population of cancer patients with small tumors and/or more than 65 years of age, suggesting that aging is no longer a limiting factor. However, aging often correlates with frailty, so the toxicity of anticancer agents is a major issue in this population. Aging by itself is also associated with cognitive dysfunction, comorbidities, and functional decline, which may have a significant impact on the patient's autonomy. Hurria outlined the state-of-the-art of cognition among elderly cancer patients. Although surprisingly only a few studies have addressed this question, some conclusions can still be drawn. Regarding specific populations, cognitive impairment is higher in women cancer survivors,[16,17] and more than 40% of elderly breast cancer patients experience objective cognitive disorders before receiving any adjuvant therapy, a proportion higher than that reported in younger patients.[18] In this respect, age as well as baseline cognitive reserve are predictors of posttreatment cognitive decline in processing speed among women treated with chemotherapy for breast cancer.[19] Finally, more than 60% of elderly breast cancer patients perceive preexisting memory problems and are more likely to perceive worsening memory after chemotherapy.[20] Patients with comorbidities may have poorer cognitive performance even before any cancer treatment. Diabetes and cardiovascular disease were associated with impairment among patients even before any adjuvant treatments.[21] Thus, including a comprehensive assessment of comorbidities (e.g., vascular risk factors, cardiovascular disease, obesity, chronic kidney disease) and comedications in cognitive studies must be encouraged for future studies.[22]

What Is the Impact of Treatments on Cognition?

The hypothesis of a dose-response relationship between chemotherapy and cognitive impairment in breast cancer patients was confirmed by the work of Collins et al.[23] After controlling for baseline performance, age, education, and mood, they found that cognitive function progressively worsened with cumulative chemotherapy exposure. Some interesting results in other types of cancers and treatments were presented during the meeting. Combination chemoradiation therapy in head and neck cancer patients also may induce cognitive dysfunction, as reported by Gan et al.[24] (these results must be tempered by the sample size, $n = 10$). In a longitudinal study presented by F. Joly, more than 30% of patients with metastatic renal cancer treated with antiangiogenic targeted therapy (mainly tyrosine kinase inhibitors) developed cognitive decline.[25]

The influence of hormone therapy was another important subject of discussion during the meeting. Despite a large number of studies having been conducted in breast cancer patients treated with hormone therapy, the clinical results are inconclusive; however, some studies suggested that tamoxifen administration may be a risk for cognitive impairment, particularly in older women.[26] Although objective cognitive functions (composite score) were shown to improve one year after cessation of adjuvant endocrine therapy in the BIG 1-98 randomized trial performed in localized breast cancer patients,[27] improvement in subjective cognitive functions was less clear.[28] Selective estrogen receptor modulators seemed to more consistently adversely affect cognitive functioning compared to aromatase inhibitors.[29] No significant longitudinal decline in cognitive performance was observed in a group of elderly breast cancer patients treated with aromatase inhibitors, even though some changes in cerebral metabolic activity (PET scan), mostly in the medial temporal lobes, were observed.[30] Discussion on the impact of anti–androgen deprivation therapy (ADT) on cognitive function in prostate cancer patients was based on the contradictory results reported in a number of small studies. Most studies did not provide substantial evidence of cognitive impairment with ADT. However, there were major methodological limitations, including the omission of cognitive domains that are theoretically at the highest risk of being affected by ADT.[31] A recent meta-analysis concluded that ADT had a significant impact on visuomotor tasks, which is consistent with the known effects of testosterone on cognitive functioning in healthy men.[32] Also, few studies discussed the impact of hematopoietic stem cell transplantation (HCT). Some neurobehavioral problems, particularly apathy, were observed in patients after HCT.[33] Although cognitive impairments are evident for a subset of patients before HCT, results from a meta-analysis suggest that cognition does not significantly change after HCT.[34]

What Is the Impact of Cancer on Cognition?

Besides exposure to cytotoxic drugs, other factors including persistent effects of surgical operation and anesthesia, psychological distress related with the diagnosis, fatigue, sleep problems, genetic factors, and tumor-related factors are involved,[35,36] suggesting an impact of cancer as a whole on cognitive function. Longitudinal studies revealed that about 20%−30% of breast cancer patients have cognitive impairment before starting adjuvant treatment.[37] These impairments seem to be more frequent in elderly patients than in younger[18] and were associated with comorbidities.[21]

J. Vardy presented results of an Australian and Canadian cohort of patients with localized colon cancer after diagnosis compared with a group of healthy controls. Before any adjuvant chemotherapy, cancer patients experienced significantly more objective cognitive impairment than healthy controls (complex processing speed, attention/working memory, and verbal learning efficiency being most affected), which supports the idea that colorectal cancer is involved per se in cognitive impairments.[38] Evaluation of cognitive functioning before treatment is essential to take into account the specific impact of treatments on cognition.

Can We Better Understand the Mechanisms Involved?

Some interesting presentations confirmed the key role of the inflammatory environment in the development of cognitive impairments. Ganz et al.[39] showed a relationship between self-reported memory complaints and some cytokine markers, including tumor necrosis factor α (TNF-α). Merriman et al.[40] reported preliminary evidence of a genetic association between an interleukin 6 (IL-6) promoter and severity of self-reported attention. This could represent a potential marker to identify patients at risk of developing attentional changes. Genetic factors such as the allele ε4 of the apolipoprotein E (APOE-ε4) also were implicated as risk factors for chemotherapy-induced cognitive decline among breast cancer survivors.[41] More recently, polymorphisms in genes related to oxidative stress/neuroinflammation, that is, genes encoding the Val allele of catechol-o-methyl transferase (COMT), which is known to influence the metabolic breakdown of catecholamines, the endothelial nitric oxide synthase (NOS3) and the prostaglandin transporter (SLCO2A1) have been shown to be associated with impaired neurocognitive functions after chemotherapy for childhood acute lymphoblastic leukemia.[42] In the Ahles et al.[43] study, patients with APOE4+ breast cancer being treated with chemotherapy who had never smoked exhibited a decline of their performance in terms of processing speed compared with patients reporting a smoking history, suggesting that smoking may correct a deficit in nicotinic receptor functioning and dopamine levels in APOE4+ individuals.

Other multifactorial parameters of cognitive impairment also were discussed. In addition to physiological toxic effects of chemotherapy, combinations of biological and medical factors such as side effects of surgical experience and anesthesia, fatigue, psychological factors or hormonal changes, and hormone-based therapy, also could play a role. Schagen et al.[44] showed that patients who were treated with chemotherapy and were informed that the treatment could affect cognition, reported higher levels of cognitive complaints. Giffard et al.[45] reported that because of the life-threatening context of the illness, cancer and its treatments also may involve many stressful events and psychiatric symptoms, such as those reported in post-traumatic stress disorder or in depression. These symptoms are known to induce adverse effects on cognition, and the autobiographical memory (i.e., memory for personal life events that gives a feeling of identity) represents one aspect of cognition that is particularly impacted by stress.[46] A relationship between hippocampal atrophy and deficit in episodic autobiographical memory retrieval was suggested, which could reflect the effect of exposure to negative events per se with heavy or cumulative stress.[47]

What Can We Learn From Neuroimaging Studies?

Significant progress has been made in neuroimaging research on cognitive functions (mainly in breast cancer) over the past few years.[48] Recent improvements and emerging data from neuroimaging studies were presented by De Ruiter and Silverman. They highlighted three important methodological aspects: 1) the trend toward the inclusion of a multimodal neuroimaging approach; 2) the conduction of prospective longitudinal studies with prechemotherapy and postchemotherapy neuroimaging according to the ICCTF recommendations,[7] and 3) the assessment of time effects since chemotherapy. The results confirmed those of previous imaging studies showing that chemotherapy is associated with structural and functional changes, notably in frontal regions involved in executive and memory processes.[49−51] Furthermore, McDonald et al.[51,52] reported on the first longitudinal prospective voxel-based morphometry (VBM) and functional MRI studies, including a large cohort of breast cancer patients. They showed that gray matter in frontal regions was reduced one month after chemotherapy completion compared with baseline and was associated with higher levels of cognitive complaints.[50] The results confirmed that this structural change was accompanied by an altered activation while performing a working memory task.[52] Patients treated with HCT also may be at risk for developing structural alterations (reduction in dorsolateral prefrontal cortex volume) as presented by Correa et al.[53]

Although changes in the microstructure of cerebral white matter had been previously suggested, Deprez et al.[54] presented for the first time a longitudinal prospective study using diffusion tensor imaging (DTI) in patients with breast cancer. She showed that three to four months after chemotherapy, fractional anisotropy (FA) values decreased notably in the corona radiata and corpus callosum, and in frontoparietal and occipital white matter tracts, with significant correlations reported between regional FA changes and attention and verbal memory scores.

Koppelmans et al.[55] explored the long-term effects of chemotherapy on brain structure in a large cohort of breast cancer survivors examined more than 20 years after chemotherapy, and showed that total brain and global gray matter volumes were smaller than in healthy controls. In contrast, VBM analyses did not reveal any focal reductions, and no significant differences were found in hippocampal volume and in white matter volume or integrity. However, FA was negatively correlated with time since chemotherapy. The authors suggested that the smaller gray matter volume observed in chemotherapy-exposed survivors might explain their worse cognitive performance. The impact of time since chemotherapy on cognition also was addressed by Conroy et al.[56] through a multimodal MRI study with VBM and functional MRI (fMRI) brain activation during a three-back task in breast cancer survivors. This is the first study to show structural and functional effects of postchemotherapy interval and to relate oxidative DNA damage to brain alterations in breast cancer survivors.

To understand the organization of whole-brain networks and to identify potential alterations related to the default mode network, Bruno et al.[57] used resting state fMRI in breast cancer survivors. Compared with healthy controls, the cancer group exhibited altered network organization including decreased global clustering values, reduced nodal degree, and hubs in areas notably implicated in executive control and memory networks. Cognitive dysfunction, therefore, may be linked to a disrupted coordination between cerebral regions in the global functional brain network. The network alteration would be indicative of a "disconnection syndrome" with lower white matter integrity.

Silverman et al.[58] explored the impact of chemotherapy on resting metabolism in breast cancer patients using fluorodeoxyglucose-positron emission tomography (FDG-PET) scans and found that metabolism in the medial prefrontal and anterior temporal cortex correlated positively with both memory complaints and cytokine levels. However, this association was only observed in patients receiving chemotherapy, suggesting that increased resting metabolism might be a compensatory mechanism for diminished functions sustained by other regions.[58]

Overall, the decrease in gray matter volume and white matter density, as well as changes in patterns of activation and in resting state metabolism, was observed several months to years after chemotherapy and was generally associated with cognitive dysfunction or complaints. Studies presented during the meeting supported several hypothesized mechanisms that may induce neuroimaging changes, notably oxidative DNA damage[56] and inflammatory processes.[58]

What Can We Learn From Animal Models Regarding the Impact of Chemotherapy, Targeted Therapies, Cancer, and Neuroprotectants on Brain Function?

Since the previous ICCTF meetings, basic science and translational research using animal models has increased. This is particularly useful to evaluate the direct impact of anticancer agents on cognitive function as well as the underlying neurobiological mechanisms, the influence of emotion and aging, and the impact of the cancer itself and potentially neuroprotective and/or preventive co-administered therapeutics.

Regarding the various antitumoral agents assessed, important data on the impact of 5-fluorouracil (5-FU), one of the major chemotherapy agents administered to colorectal and breast cancer patients, on cognitive function were presented. H. Castel reported a study showing that more than seven weeks after the end of treatment, 5-FU induced an alteration of cognitive flexibility and behavioral hyperreactivity to novelty, indicating that long-term chemotherapy alters activity of selective brain areas connected to hippocampo-cortico-fronto-cerebellar networks (in the dentate gyrus of the hippocampus).[59] It also has been reported by A. Chambers that 5-FU and leucovorin administration in rats led to a decreased survival of newly dividing cell survival in the subgranular zone of the dentate gyrus immediately after treatment, which continued to decline over the following six weeks. This mechanism would be involved in the cognitive impairments in rats described by the same team after 5-FU treatment.[60,61] In the study presented by Christie et al.,[62] chronic treatment of nude rats with cyclophosphamide or doxorubicin induced significantly altered performance on both the novel location recognition task and the contextual fear conditioning task compared to untreated groups, suggesting a disruption of hippocampal-based memory functions. In addition, Dietrich et al.[63] suggested that corticosteroids, which have been widely used for decades in cancer patients, could alter oligodendroglial cells. A study by Seigers et al.,[64] which individually tested chemotherapeutic compounds, allowed multiple comparisons in terms of schedule administration, dose, and time after cognitive investigations. Overall, 5-FU and other cytotoxic drugs currently used in the treatment of the most prevalent cancers were shown to impact spatial memory and/or behavioral flexibility and to be associated with inhibition of proliferating neural precursors and/or mature neuron integrity.

It is unclear whether the lower cognitive reserve of the elderly or the immaturity of juvenile brains constitutes a risk factor for chemotherapy-induced cognitive deficits. By using a mouse model, Dubois et al. showed that the poor performances of 5-FU-treated animals were not worsened by age (with no impairment of executive functions).[59]

The alteration of executive functions was not observed when 5-FU was administered with glucose, and glucose also prevented the long-term 5-FU-induced decrease in hippocampal cell proliferation.[59] Winocur et al.[65] demonstrated that a combination of 5-FU and methotrexate administered to adult mice led to a poorer memory for tasks acquired before treatment, as well as an altered learning and memory for acquired tasks.

Animal models allowed the evaluation of the impact of targeted therapies, a class of agents that may permeate the blood-brain barrier and act directly in the brain, thus affecting cerebral angiogenesis and functioning.[66] Dubois et al. established that mice treated with the mammalian target of rapamycin (mTOR) inhibitor everolimus displayed a marked reduction in weight gain that persisted after the end of the treatment period, accompanied by an altered cytochrome oxidase activity in selective cerebral regions involved in energy balance, food intake, reward, learning and memory modulation, sleep/wake cycle regulation, and arousal. However, everolimus did not alter emotional reactivity, learning and memory performances, behavioral flexibility, or reactivity to novelty. Thus, everolimus caused selective metabolic modifications in cerebral regions under control of the hypothalamus, but with no impact on cognition.[67]

What Can Be Expected From Cognitive Rehabilitation in Cancer Survivors?

How to optimally help patients cope with cognitive complaints and dysfunctions is an emerging area of research in oncology. For most patients, the effects of cancer and its treatment on cognitive function are transient, but for others, the symptoms persist and have a deleterious impact on their quality of life and daily activities.[68,69] For the first time, some interventional studies were presented during the meeting.

Schuurs and Green[70] tested the feasibility and effectiveness of a four week cognitive rehabilitation approach aimed at improving cognitive function and quality of life in patients having completed cancer treatments. The authors compared three groups of patients: a group of non-CNS cancer survivors with cognitive support and two groups without cognitive rehabilitation (a group of cancer survivors and a group of adults without cancer). The intervention was effective in improving overall cognitive function, visuospatial constructional performance, and delayed memory beyond practice effects alone. It also was helpful in reducing participants' perceptions of cognitive impairment and psychosocial distress, as well as in promoting social functioning and their understanding of cognition. The improvements were maintained over the three months after the intervention. A double-blind, randomized, wait-list controlled study assessing the effectiveness of cognitive rehabilitation in cancer survivors[71] was presented. The intervention consisted of a seven week cognitive rehabilitation program delivered in group format. All participants reported subjective cognitive complaints. Cancer survivors in the rehabilitation group experienced a significant improvement in memory and attention task, but other abilities such as executive functions or spatial abilities were not modified.

Ercoli et al.[72] presented a cognitive rehabilitation pilot program for breast cancer survivors comprising five once-weekly, two-hour sessions including psychoeducation, in-class and at home cognitive exercises, and goal setting. After the rehabilitation intervention, motor and psychomotor speed tests and the Patient Assessment of Own Functioning Inventory (PAOFI) differed significantly from baseline. All cognitive functions, except motor speed, remained significant two and four months after the intervention. Taken together, these results suggest that cognitive rehabilitation may be an effective intervention for treating cognitive dysfunction in cancer patients.

Perspectives Emerging From the Paris Meeting

Clinical and Neuropsychological Updated Research

The impact of cancer and its treatments on cognition is an important subject of debate in the oncology community as suggested by the high number of recent publications (more than 45 reviews published since our meeting in 2012). The neuro-oncological research community has become more aware of the importance of cognitive function as an outcome; today, they include cognitive assessment with standardized neuropsychological tests in most of their clinical trials in CNS patients.[73]

Two meta-analyses have been published in adult patients with non-CNS cancers. In the meta-analysis by Jim et al.,[2] dealing with breast cancer patients previously treated with standard dose chemotherapy, results indicate that, on average, observed cognitive deficits are small in magnitude and limited to the domains of verbal ability and visuospatial ability. Hodgson et al.[74] examined the effects of chemotherapy on seven cognitive domains across five cancer types. They found evidence of the presence of cognitive impairment for executive function and memory. No relationship was found between cognitive impairment and time since treatment cessation, but a significant negative relationship was found for treatment duration. Age seemed to have no impact on treatment-related cognitive impairment, but most of the patients included in the meta-analysis were younger than 60 years.

Large studies with long-term follow-up have been conducted among survivors of pediatric brain tumors. Survivors treated with cranial radiotherapy and/or chemotherapy showed seriously impaired intellectual functioning and attentiveness.[75] Similar results were observed among children treated with intrathecal methotrexate for leukemia.[76] Young adult long-term survivors of childhood Hodgkin disease are also at risk for neurocognitive impairment.[77] There are some recent advances in interventions for neuropsychological sequelae including pharmacologic treatments, cognitive remediation derived from traumatic brain injury rehabilitation, and comprehensive school reintegration efforts. In pediatric survivors, the use of stimulant medication such as methylphenidate has provided promising outcomes.[78]

Several recent clinical studies suggest that other cancer populations treated with chemotherapy, that is, testicular cancer and colon cancer, may experience similar cognitive symptoms. A recent study in patients with testicular cancer found that compared with a surveillance group, patients treated with BEP (bleomycin, etoposide, cisplatin) chemotherapy had high rates of cognitive decline at 12 months after treatment.[79] Similarly, a prospective study in patients with colon cancer receiving oxaliplatin plus 5-fluorouracil/leucovorin (FOLFOX4) revealed adverse effects on verbal memory in a subgroup of patients.[80]

Other cancer treatments may induce cognitive symptoms. Emerging data suggest that vascular endothelial growth factor (VEGF) plays a role in brain cognition and that VEGF inhibitors may induce neurotoxic effects on a patient's cognitive function.[81] A case-control study in patients with metastatic renal cell cancer and gastrointestinal stromal tumors suggested that anti-VEGF tyrosine kinase inhibitors may induce a deleterious impact specifically on executive functions and learning and memory.[82]

Different interventional strategies have been explored in the last two years: Mindfulness-based stress reduction and physical exercise or mental stimulation have been gaining increasing attention.[73] Preliminary results of adaptation training suggest that this is safe and feasible.[83] Cognitive training, used in breast cancer survivors with a novel, online, home-based program targeting executive function, significantly improved multiple executive function skills reported by objective and self-report measures.[84] The effect of exercise (like qigong) was evaluated in longitudinal studies and preliminary results demonstrated significant improvements in subjective cognitive function tests as well as on psychological and physical health.[85] Pharmacological interventions also are being tested. Modafinil and methylphenidate have been investigated, yielding conflicting results, and further research is ongoing.[86]

Translational Research

With advances in translational research, some mechanisms implicated in cognitive disorders are becoming better understood. Since the ICCTF Paris meeting, it has been shown in animal models that frontal cortical biochemical mechanisms are involved in cognitive disorders.[59,87] The finding of specific hippocampo-cortico-prefrontal circuit and behavioral flexibility altered by chemotherapy also was reported by Sharpe et al.[88] in a model evaluating oxaliplatin. Some studies focused on cerebral fatigue, which is often associated with cognitive complaints. An original study investigated to what extent chemotherapy (cyclophosphamide-doxorubicin-5-FU) may induce inflammatory cytokine brain expression and an alteration of physical activity in rats. The data showed that chemotherapy provokes inflammatory gene expression in the brain stem and hypothalamus, and suppresses hypothalamic orexin neuron activity. Central injection of orexin A restored the ambulatory activity in the animals.[89] As a potential way of intervention, mild exercise has been shown to be beneficial in chemotherapy-induced cognitive deficits in rats.[90] In addition, forced running exercise for rats has been found to prevent radiation-induced cognitive deficits and to partially compensate for the drastic decrease in hippocampal proliferating cells and of the brain-derived neurotrophic factor-phospho-cAMP response element-binding protein (BDNF-phospho-CREB) pathway.[91] It has also been shown that intrathecal administration of memantine, an N-methyl-D-asparte (NMDA) receptor antagonist, reduces the incidence of methotrexate-induced cognitive deficits and may be of benefit for cancer patients receiving methotrexate.[92] Finally, it was proposed that high levels of the antioxidant enzyme superoxide dismutase could provide protection against radiation-related defects in hippocampal functions.[93]

Some new studies have investigated the biological effect of the tumor itself on the pathogenesis of cancer-associated brain disorders. In the study by Yang et al.,[94] tumor-bearing mice exhibited depressive behavior and impaired object recognition memory. This was associated with hippocampal reduced levels of messenger RNA encoding brain-derived neurotrophic factor, cyclooxygenase 2, IL-6, and tumor necrosis factor α (TNF-α), and of proliferation of neuronal progenitors through modification of the hypothalamo-pituitary-adrenal axis.

Overall, these important studies published between 2012 and 2014 highlight the major contribution of animal models in addressing the mechanisms of "cancer-fog," cancer-related cognitive impairment, and suggest new interventional targets for clinical trials.

Imaging Studies

Imaging studies published since the ICCTF 2012 conference are often comprehensive, with the inclusion of various neuroimaging variables (i.e., multimodal MRI), the evaluation of self-report measures, neurocognitive tests and blood draws. Sample sizes seem to be increasing, and there is a tendency to examine associations of brain imaging with behavior and underlying mechanisms.

Volumetric studies have evaluated neurotoxic side effects of systemic cancer treatment in patients with hematologic and breast cancers.[53,95] These studies confirmed the idea that anterior portions of the brain are more vulnerable to adverse effects of chemotherapy than posterior regions. Blood draws indicated that cytokine levels (TNF-α and IL-6) were elevated in comparison to the healthy controls.[95] Hippocampal volume showed complex associations with cytokine levels, possibly indicating that hippocampal mechanisms might be altered in the context of elevated cytokine levels. Interestingly, longer time since treatment, but not earlier disease stage, was associated with lower levels of TNF-α, suggesting that cancer treatment, rather than the disease itself, determines TNF-α levels and that TNF-α remained elevated after chemotherapy ($\pm$5 years) but did further decrease with time.

At least nine novel functional fMRI studies have been published since the ICCTF Paris meeting.[56,57,96–102] Except for one study in prostate cancer patients,[98] all investigations were conducted in breast cancer patients/survivors. Worry associated with reduced demand-related deactivation in default mode regions appears to be an important contributor to neurocognitive dysfunction, independent of adjuvant treatment for breast cancer.[96] Disrupted functional connectivity patterns of default mode network connectivity seems to be specific to chemotherapy exposure among breast cancer survivors and may help explain long-term cognitive sequelae as suggested by Kesler and Hosseini.[100]

Several PET studies were published since the ICCTF meeting.[58,103–105] D'Agata et al.[103] explored differences of metabolic glucose rate through the brain resting state FDG-PET between chemotherapy-treated and nontreated patients with lymphoma and showed clear-cut patterns that discriminate the two groups, notably a hypometabolic network in chemotherapy patients particularly in prefrontal cortex and cerebellar areas. This finding could be compared with those of Bruno et al.[57] who found, in chemotherapy-treated breast cancer survivors, a disrupted network in frontal, striatal, and temporal areas.

Conclusions

Research on the impact of treatment and cancer on cognitive function has been growing over the past few years, with a multidisciplinary approach. The methodology and the design of the clinical studies have dramatically improved.

The most recent data, mainly obtained in patients with breast cancer, support the concept that non-CNS cancer per se may be involved in cognitive dysfunction. Some argument suggested that parameters of inflammation may be involved. There are emerging data on cognitive dysfunction in patients treated with chemotherapy for various types of cancers (not only breast cancer) or with targeted therapies, hormone therapy, ADT, and HCT. Large longitudinal studies in these groups of patients should be encouraged. We also need to improve our knowledge regarding the influence of comorbidities and aging to better understand which factors are implicated in the development of cognitive dysfunction.

Animal models confirmed that chemotherapy directly induces objective cognitive damage. New preclinical exploratory studies are needed to specify to what extent the action of targeted therapies is similar. Finally, functional neuroimaging focused on objective and subjective cognitive dysfunction and suggested an associated decrease in gray and white matter and changes in patterns of activation.

Some interesting perspectives regarding the effect of cognitive rehabilitation are currently under evaluation, which may help patients cope with this troublesome side effect.

Disclosures and Acknowledgments

This research received no specific funding/grant from any funding agency in the public, commercial, or not-for-profit sectors. The authors declare no conflicts of interest.

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